Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Clusters of post-acute COVID-19 symptoms: a latent class analysis across 9 databases and 7 countries

Daniel Prieto-Alhambra

daniel.prietoalhambra@ndorms.ox.ac.uk

Centre for Statistics in Medicine, NDORMS, University of Oxford, Oxford, UK https://orcid.org/0000-0002-3950-6346

Kim López Güell

https://orcid.org/0000-0002-8462-8668

Martí Català

University of Oxford

Daniel Dedman

CPRD, Medicines and Healthcare products Regulatory Agency, London, UK https://orcid.org/0000-0002-3699-5391

Talita Duarte-Salles

Fundació Institut Universitari per a la recerca a l'Atenció Primària de Salut Jordi Gol i Gurina (IDIAPJGol), Barcelona, Spain.

Raivo Kolde

Institute of Computer Science, University of Tartu, Estonia

Raúl López-Blasco

Aragon Health Sciences Institute (IACS), Zaragoza, Spain https://orcid.org/0000-0002-1909-9931

Álvaro Martínez

The Health Research Institute Hospital La Fe, Avenida Fernando Abril Martorell, 106 Torre A 7a planta, 46026, Valencia, Spain

Gregoire Mercier

Public Health Department, University Hospital of Montpellier, 34295 Montpellier, France

Alicia Abellan

Fundació Institut Universitari per a la recerca a l'Atenció Primària de Salut Jordi Gol i Gurina (IDIAPJGol), Barcelona, Spain

Johnmary Arinze

Department of Medical Informatics, Erasmus University Medical Center, Rotterdam, the Netherlands

Theresa Burkard

Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences, University of Oxford, Oxford, UK

Edward Burn

Fundació Institut Universitari per a la recerca a l'Atenció Primària de Salut Jordi Gol i Gurina (IDIAPJGol), Barcelona, Spain, https://orcid.org/0000-0002-9286-1128

Zara Cuccu

CPRD, Medicines and Healthcare products Regulatory Agency, London, UK

Antonella Delmestri

University of Oxford https://orcid.org/0000-0003-0388-3403

Dominique Delseny

Public Health Department, University Hospital of Montpellier, 34295 Montpellier, France

Sara Khalid

University of Oxford https://orcid.org/0000-0002-2845-5731

Chungsoo Kim

Department of Biomedical Sciences, Ajou University Graduate School of Medicine, Suwon, Republic of Korea

Ji-woo Kim

Big Data Department, Health Insurance Review and Assessment Service, Wonju, Republic of Korea

Kristin Kostka

Pharmaco- and Device Epidemiology Group, CSM, NDORMS, University of Oxford

Cora Loste

Department of Infectious Diseases & irsiCaixa AIDS Research Institute, Hospital Universitari Germans Trias i Pujol, Badalona, Catalonia, Spain

Miguel Mayer

Hospital del Mar Medical Research Institute (IMIM) https://orcid.org/0000-0003-0362-6298

Jaime Meléndez-Cardiel

Biocomputing Unit, Aragon Health Sciences Institute (IACS), Zaragoza, Spain https://orcid.org/0000-0002-9189-8543

Nuria Mercadé-Besora

Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences, University of Oxford, Oxford, UK

Mees Mosseveld

Department of Medical Informatics, Erasmus University Medical Center, Rotterdam, the Netherlands https://orcid.org/0000-0001-9301-8565

Akihiko Nishimura

Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA https://orcid.org/0000-0002-6932-2513

Hedvig ME Nordeng

Pharmacoepidemiology and Drug Safety Research Group, Department of Pharmacy, Faculty of Mathematics and Natural Sciences, University of Oslo, Oslo, Norway

Jessie O Oyinlola

CPRD, Medicines and Healthcare products Regulatory Agency, London, UK

Roger Paredes

Hospital Universitari Germans Tries i Pujol, Case Western Reserve University School of Medicine

Laura Pérez-Crespo

Fundació Institut Universitari per a la recerca a l'Atenció Primària de Salut Jordi Gol i Gurina (IDIAPJGol), Barcelona, Spain.

Marta Pineda-Moncusí

University of Oxford https://orcid.org/0000-0003-0567-0137

Juan Manuel Ramírez-Anguita

Parc de Salut Mar, Hospital del Mar Medical Research Institute, Barcelona, Spain

Nhung TH Trinh

Pharmacoepidemiology and Drug Safety Research Group, Department of Pharmacy, Faculty of Mathematics and Natural Sciences, University of Oslo, Oslo, Norway

Anneli Uusküla

Institute of Family Medicine and Public Health, University of Tartu, Tartu, Estonia

Bernardo Valdivieso

The Health Research Institute Hospital La Fe, Avenida Fernando Abril Martorell, 106 Torre A 7a planta, 46026, Valencia, Spain

Junqing Xie

Centre for Statistics in Medicine and NIHR Biomedical Research Centre Oxford, NDORMS, University of Oxford
https://orcid.org/0000-0002-0040-0042

Lourdes Mateu

Department of Infectious Diseases & irsiCaixa AIDS Research Institute, Hospital Universitari Germans Trias i Pujol, Badalona, Catalonia, Spain

Annika Jödicke

Centre for Statistics in Medicine and NIHR Biomedical Research Centre Oxford, NDORMS, University of Oxford, Oxford, UK

Article

Keywords:

Posted Date: April 10th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4091654/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: Yes there is potential Competing Interest. D.P.A.'s department has received grant/s from Amgen, Chiesi–Taylor, Lilly, Janssen, Novartis, and UCB Biopharma. His research group has received consultancy fees from Astra Zeneca and UCB Biopharma. Amgen, Astellas, Janssen, Synapse Management Partners and UCB Biopharma have funded or supported training programmes organised by DPA's department. R.P. reports serving on advisory boards for Gilead Sciences, Inc, Pfizer, Inc, Roche Therapeutics, MSD, GSK, ViiV Healthcare, Eli Lilly and Company, PharmaMar, and Atea Pharmaceuticals, Inc; and receiving research grants paid to his institution from MSD, ViiV Healthcare, Gilead Sciences, and PharmaMar. L.M. reports receiving grants from Grifols; receiving honoraria as a speaker from AstraZeneca, Gilead Sciences, GSK, and Pfizer; and participation in advisory boards for Gilead Sciences and Merck. M.M. works for a research group that in the past 3 years received unconditional research grants from Chiesi, UCB, Amgen, Johnson & Johnson, Innovative Medicines Initiative and the European Medicines Agency. D.Ded., Z.C. and J.O. are employees of the Medicines and Healthcare Products Regulatory Agency, which provides the CPRD research service. K.K. is a consortial author in the US National Institutes of Health National COVID Cohort Collaborative (funding expired in 2022 with no renewal or active impact on any current work).

Abstract

Prior evidence has suggested the multisystem symptomatic manifestations of post-acute COVID-19 condition (PCC). Here we conducted a network cluster analysis of 24 WHO proposed symptoms to identify potential latent subclasses of PCC. Individuals with a positive test of or diagnosed with SARS-CoV-2 after 09/2020 and with at least one symptom within ≥ 90 to 365 days following infection were included. Sub-analyses were conducted among people with ≥ 3 different symptoms. Summary characteristics were provided for each cluster. All analyses were conducted separately in 9 databases from 7 countries, including data from primary care, hospitals, national health claims and national health registries, allowing to validate clusters across the different healthcare settings.

787,078 persons with PCC were included. Single-symptom clusters were common across all databases, particularly for joint pain, anxiety, depression and allergy. Complex clusters included anxiety-depression and abdominal-gastrointestinal symptoms.

Substantial heterogeneity within and between PCC clusters was seen across healthcare settings. Current definitions of PCC should be critically reviewed to reflect this variety in clinical presentation.

Introduction

With more than 200 post-acute COVID-19 symptoms being reported in the context of “long COVID”, the clinical presentation of PCC is highly variable among affected individuals[1–6]. While studies estimate that at least around one in ten[7] people with COVID-19 develop post-acute symptoms, this new condition is still poorly understood. However, with as many as 60 million people potentially affected worldwide, PCC presents not only a huge burden to affected individuals, but also poses enormous healthcare, economic, and socio-political challenges to the wider society[8].

To enable targeted clinical care and facilitate research, clinical experts selected a list of 25 relevant symptoms during a Delphi process to develop a clinical case definition of PCC, subsequently published by the World Health Organization (WHO)[9].

Machine learning methods including clustering techniques have been used to identify patterns of associated symptoms in the approach to identify clinical subgroups of PCC[10–13], which can subsequently facilitate targeted research into potential treatments and optimized care for affected individuals. However, previous studies using clustering approaches utilized small, prospectively collected datasets such as self-reported surveys and are therefore often not representative of people with PCC[10–13]. Moreover, the methodological heterogeneity of these studies in terms of setting, design, symptom definition, and statistical approach prevents general comparison of clusters across studies. However, recent studies have illustrated the validity and relevance of real-world data to study PCC in large populations[14].

Our study used multiple real world data assets from across Europe and North America to identify clusters of PCC and to validate them across different geographies and healthcare settings.

Results

Study population characteristics

A total of 787,078 individuals with PCC were included out of 3,209,411 eligible COVID-19 participants from nine databases from six European countries and the US, with 36,779 people included from Estonia (database CORIVA), 344

from France (database eDOL), 258,995 from Norway (database NLHR@UiO), 84,284 from the Netherlands (database IPCI), 38,798 from Spain (databases HULAFE and IMASIS), 234,736 from the United Kingdom (databases CPRD GOLD and CPRD AURUM), and 133,675 from the United States (database Pharmetrics® Plus for Academics). Table 1 summarises participant characteristics in each database. Participants with PCC were predominantly female (except in eDOL), and of median age between 35 and 52 years old, except for hospital databases IMASIS and eDOL where median age was ≥ 60 years old. Most participants only had one post-acute COVID-19 symptom recorded, whereas 21–51% of participants had at least two symptoms and 4–25% had at least three symptoms.

Table 1
Summary characteristics of people with PCC

Database name	Country, region	Healthcare setting	N	≥ 2 symptoms	≥ 3 symptoms	Median age	Sd age	Sex female
CORIVA	Estonia, nationwide	Claims, prescriptions	36779	28.3%	7.6%	42.9	13.6	23747 (64.6%)
CPRD Aurum	UK, nationwide	Primary care	217236	28.3%	7.8%	40.4	19.9	139467 (64.2%)
CPRD GOLD	UK, nationwide	Primary care	17500	23.7%	5.8%	44.4	14.8	11371 (65.0%)
eDOL	France, Montpellier	Secondary care	344	NA	NA	62.2	24.4	145 (42.0%)
HULAFE	Spain, Valencia	Secondary care	35587	35.9%	12.1%	42.4	15.3	22159 (62.3%)
IMASIS	Spain, Barcelona	Secondary care	3211	28.4%	NA	61.0	12.7	1698 (52.9%)
IPCI	Netherlands, nationwide	Primary care	84284	21.4%	4.3%	40.4	13.6	52864 (62.7%)
NLHR@UiO	Norway, nationwide	National health Registries (primary secondary care)	258995	24.1%	5.6%	35.1	19.1	154688 (59.7%)
Pharmetrics® Plus for Academics	USA, nationwide	Claims	133675	51.0%	25.2%	52.2	13.1	78725 (58.9%)

Table 1. Clinical characteristics and demographics of study population by database. CPRD stands for Clinical Practice Research Datalink, IPCI for Integrated Primary Care Information, HULAFE for The Health Research Institute Hospital La Fe, IMASIS for Parc de Salut Mar Barcelona Information System. NA is for non-available data.

Proportions of individual symptoms in people with at least one recorded symptom (Fig. 1), and in a subgroup of people with at least two and at least three different symptoms (Supplementary Fig. 2) were estimated in all databases. The OMOP Extension concept identifier is 705076, which accounts for ‘Post-acute COVID-19" condition diagnosis, was included as an additional covariate to explore its correlation with individual symptoms.

While the most common symptoms were generally different across databases, consistently common symptoms included abdominal pain, anxiety, gastrointestinal issues or joint pain.

Clustering

Participants with PCC were classified into different groups based on their patterns of associated symptoms through Latent Class Analysis (LCA). Clustering analyses were conducted first among people with more than one recorded symptom, and then in subgroups of people with at least two and at least three different symptoms, to narrow the definition of post-acute COVID-19.

Models with four Latent Class clusters represented the best fit in seven out of nine databases. Supplementary Table 4 shows the “best cluster model” (with a specific number of classes) according to different information criteria statistics, for the cohorts of individuals with at least one, two or three symptoms. After discussing the results with expert clinicians, we concluded that the clinical interpretation for three clusters was very similar to that of four clusters, and we therefore report here results of three clusters for all databases. Results for four clusters (and for two, five, six and seven) are available in the interactive web application: https://dpa-pde-oxford.shinyapps.io/WP3_Clustering/.

Clustering among people with PCC

Results for the clusters among people with one or more post-acute symptoms are presented in Fig. 2. The full legend can be seen in Supplementary Fig. 3. With most people having only one recorded post-acute COVID-19 symptom, many of the clusters were characterised by single symptoms or at most a combination of two highly prevalent symptoms. Single-symptom clusters defined by highly prevalent symptoms were seen across multiple databases, including “joint pain” in NLHR@UiO and CPRD GOLD, and “allergy” in IMASIS, NLHR@UiO and CORIVA.

An “anxiety, depression” cluster was the only multi-symptom cluster seen consistently across more than one database (CPRD AURUM and eDOL). Another multi-symptom cluster included the combination of “dyspnoea, chest pain, fatigue, GI symptoms, tachycardia”, but this was only seen in US health claims (PharMetrics® Plus for Academics).

Clustering among people with at least three symptoms

When restricting to the subgroup of people with at least three symptoms, eDOL and IMASIS did not have the required more than five hundred individuals to contribute to the analyses.

Results from clustering among people with at least three different symptoms are illustrated in Fig. 3. The full legend can be seen in Supplementary Fig. 3. Clusters combining “anxiety, depression”, “abdominal pain, gastrointestinal issues” and “cough, dyspnoea, fatigue, joint pain, chest pain” were identified across almost all databases. Table 2 summarises multi-symptom clusters with their baseline characteristics.

The cluster characterised by "anxiety, depression” had the highest percentage of women, was seen most often in the 3rd decade of life, and was sometimes accompanied by additional symptoms like headache, joint pain (HULAFE), fatigue malaise and headache (IPCI) and fatigue malaise, headache and allergy (NLHR@UiO).

The "abdominal pain, gastrointestinal issues” cluster was seen most often in the 3rd and 4th decades of life and accompanied by additional symptoms such as intermittent fever, cough and allergy in one case (HULAFE).

The last cluster, characterised by "cough, dyspnoea, fatigue malaise, joint pain, chest pain”, was the one with the highest percentage of males, most common in patients in their 5th or 6th decade of life, and accompanied by tachycardia in Pharmetrics® Plus for Academics. Additionally, the OMOP Extension “diagnosis” code for post-COVID-19 condition was associated with this cluster in CPRD AURUM.

A description of the different clusters per database for people with at least three symptoms can be found in Supplementary Table 5. Analyses among people with at least two different symptoms are available in the Supplementary Fig. 4. All results are available in an interactive web application [https://dpa-pde-oxford.shinyapps.io/WP3_Clustering/].

Table 2. Summary of the three-cluster solution for the ≥3 symptoms cohorts across contributing databases. In green, clusters found in the respective database. In orange otherwise. CPRD stands for Clinical Practice Research Datalink, IPCI for Integrated Primary Care Information, HULAFE for Hospital Universitario La Fe, IMASIS for Parc de Salut Mar Barcelona Information System and NLHR@UiO stands for Norwegian Linked Health Registries at University of Oslo.

CLUSTER DESCRIPTION	CORIVA	CPRD Aurum	CPRD GOLD	HULAFE	IPCI	Pharmetrics® Plus for Academics	NLHR@UiO
Depression Anxiety	**N** 530 (19%) **Age** 37 (sd 15) **Male** 13%	**N** 5056 (30%) **Age** 36 (sd 16) **Male** 23%	**N** 334 (33%) **Age** 38 (sd 15) **Male** 27%	**N** 2182 (51%) **Age** 47 (sd 13) **Male** 20% *+ headache, joint pain*	**N** 1396 (39%) **Age** 35 (sd 15) **Male** 23% *+ fatigue malaise, headache*	These two clusters are approximately merged in this database.	**N** 4197 (29%) **Age** 34 (sd 13) **Male** 27% *+ fatigue malaise, headache, allergy*
Gastrointestinal issues Abdominal pain		**N** 4382 (26%) **Age** 40 (sd 19) **Male** 27%	**N** 263 (26%) **Age** 47 (sd 19) **Male** 24%	**N** 816 (19%) **Age** 19 (sd 13) **Male** 38% *+ intermittent fever, cough, allergy*	**N** 1324 (37%) **Age** 37 (sd 17) **Male** 24%		**N** 5091 (35%) **Age** 26 (sd 12) **Male** 22%
Cough Dyspnoea Fatigue Joint pain Chest pain		**N** 7415 (44%) **Age** 53 (sd 19) **Male** 32% *+* OMOP Extension *post-COVID-19 code*	**N** 416 (41%) **Age** 58 (sd 20) **Male** 30%			**N** 8425 (25%) **Age** 61 (sd 15) **Male** 48% *+ tachycardia*	**N** 5246 (36%) **Age** 54 (sd 15) **Male** 44%

Discussion

Summary of key results

This is to our knowledge the first and largest international study reported to date on post COVID-19 symptoms. We identified different patterns of symptoms associated with PCC using latent class models, and subsequently characterised the derived clusters across a large network of nine different databases from seven countries and over 700,000 patients.

When clustering among people with PCC, as defined by WHO, those clusters were dominated by single isolated symptoms, e.g., joint pain, or clinically related symptoms such as anxiety and depression. These are prevalent in the general population and might not necessarily be characteristic or differential for PCC.

Clusters identified among people with multiple symptoms, however, showed substantially more complex multi-symptom patterns. Among those, some specific clinical subgroups were observed across multiple databases, and were characterised by mental health-related symptoms, gastrointestinal issues or respiratory symptoms, sometimes associated with fatigue. These patterns of symptoms align with the previous literature and clinical presentation of people seeking specialist care for PCC[15].

Results in context

While our study is the first to directly compare subgroups of people with PCC across different countries and healthcare settings, previous studies used clustering algorithms to understand patterns of clinically relevant symptoms in people with PCC: Gottlieb et al[10] used latent class analyses to identify phenotypes of PCC based on survey data among 4,505 people with COVID-19 in the U.S. during 3- and 6 months following their SARS-COV-2 infection. Four distinct clusters were identified: one with almost no symptoms, one with tiredness/headache/musculoskeletal issues, one with loss of taste and smell and one with “many symptoms across multiple systems”. Similar studies were conducted using survey data from prospective cohort studies among 233 and 497 people presenting at PCC clinics in Ireland and Japan. Using a hierarchical clustering approach, Kenny et al.[12] identified three clusters of symptoms dominated by pain (joint pain, myalgia, headache), cardiovascular symptoms (e.g. chest pain, shortness of breath and palpitations), and a “substantially smaller number of concomitant symptoms”, while Tsuchida and colleagues[11] described five clusters based on self-reported symptoms dominated by “fatigue”, “fatigue/cardiovascular symptoms/forgetfulness”, “fatigue/headache/mental health issues/forgetfulness”, “hair loss” and “taste and smell disorders”. Other studies included additional markers on top of persisting symptoms for their clustering approach, e.g., COVID-19 disease severity[13], comorbidity burden[13,16], symptoms at inclusion[13,16], hospitalisation data[16] and quality of life[13] to characterise clusters by the number of symptoms and disease severity[13,16], and impact on daily living activities[16].

Only a few studies, however, have assessed cluster reproducibility in the context of PCC. Reese et al.[17] used k-means clustering on data from the National COVID Cohort Collaborative (N3C) to derive clusters of people with PCC among people diagnosed with an ICD-10 diagnosis code U09.9 in six US hospitals. Clustering was first performed on the database with the largest number of patients affected by PCC, with six clusters derived: “multisystem/laboratory”, “pulmonary”, “neuropsychiatric”, “cardiovascular”, “pain/fatigue” and “Multisystem/pain”. Generalizability of clusters was subsequently assessed by assigning patients from other healthcare centres to clusters based on high semantic similarity of symptoms recorded in original patients. The identified clusters were found to be reproducible across the different hospital systems tested. Similarly, Zhang et al[18] validated phenotype subgroups among people with PCC, suggesting four clusters consistent across two US health databases: “cardiac/renal”, ”respiratory/sleep/anxiety”, “musculoskeletal/nervous system”, “digestive/respiratory system”. While these studies provide important insights in the reproducibility and robustness of derived clinical subgroups, no validation has been conducted across different countries and healthcare settings. During the pandemic healthcare policies varied substantially across countries, particularly with respect to access to care, and patient profiles may vary greatly in the hospital vs. outpatient setting.

Recently, a large meta-analysis[15] comprising 1.2 million people from 22 countries suggested that clinical symptoms for as much as 51%, 60% and 35% of people with post COVID-19 conditions can be described by at least one of three PCC clusters: persistent fatigue with body pain and mood swings, ongoing respiratory problems, or cognitive problems. However, those clusters were defined based on reporting frequency in previous studies, and not directly derived from data.

Several clusters identified in our study are consistent with previous literature, with “pain” and “mental health issues” being commonly reported, and many clusters being associated with some degree of fatigue or tiredness[19,20]. Moreover,

our sensitivity analysis including the “post-acute COVID-19 condition” diagnosis code as an additional symptom showed its association with a cluster characterised by cough, dyspnoea and fatigue in CPRD AURUM. It is likely that this relates to the most common presentation in specialised clinics.

Strengths and limitations

This study has several strengths. We conducted a multi-database, international network study including data from different healthcare settings and including a large number of people with PCC. The main differences between previous studies might be explained by the methodological heterogeneity, including study populations, settings, post-acute COVID-19 definition and clustering methods. We used a single standardized methodological approach including a consistent definition of PCC, and statistical analyses, increasing the robustness and comparability of our clusters across databases and settings. The use of a recognised common data model, OMOP CDM[21,22], allowed for federated analyses using the same code, shared in line with open science practises. This minimised further the bias in the process and maximised reproducibility. The outstanding similarity between the results of the similar CPRD GOLD and CPRD AURUM, in terms of the cluster solutions attained, databases reassure the robustness of the results. There is, however, some heterogeneity across databases, which can be explained by underlying heterogeneity between geographies and healthcare settings. A further limitation is that routinely collected data does not include information on the severity of individual symptoms. Similarly, the validity of recording of symptoms in such data is subject to misclassification and incompleteness, as information is collected in actual healthcare settings by multiple healthcare professionals. Furthermore, the motivation to report PCC might vary across settings. This makes it harder to compare clusters both within and across databases.

Implications and recommendations for future research

PCC is a debilitating illness that afflicts individuals in their 40s, incurring a substantial economic cost[23]. Healthcare systems must be well-prepared to address the needs of millions of patients. Accurately estimating the number of affected individuals is essential for effective resource planning.

While the WHO definition of PCC proves useful, it is inevitably subject to develop over time given new research findings and does not consider a minimum number of symptoms. The identification of large single-symptom clusters characterized by highly common symptoms suggests that the WHO definition of long COVID needs refining, with the inclusion of more than one co-occurring symptom.

The identification of different patterns of symptom clusters can help optimize treatment strategies across different groups, as well as aid in identifying potential diverse pathogenic mechanisms.

Conclusion

This study included more than 787,000 persons with post-acute COVID-19 condition from seven countries, and found both single and complex symptom clusters, mainly “anxiety and depression”, “gastrointestinal issues and abdominal pain”, and “cough, dyspnoea, fatigue, joint and chest pain”.

Single-symptom clusters likely reflect symptoms with high prevalence in the general population as well as common complications of COVID-19. Current definitions for PCC should therefore be critically reviewed and refined where needed to reflect the multiplicity of symptoms and variety in clinical presentations of PCC.

Methods

Data sources

This multinational study was based on routinely collected, de-identified healthcare records from seven countries and nine databases, covering a broad range of healthcare settings. Primary care records were retrieved from the Netherlands (Integrated Primary Care Information database, IPCI[24]) and the UK (Clinical Practice Research Datalink CPRD GOLD[25] and CPRD Aurum[26]). Hospital records were analysed from the The Health Research Institute Hospital La Fe (HULAFE, Valencia, Spain), the Parc de Salut Mar Barcelona Information System (IMASIS[27], Barcelona, Spain) and the Centre Hospitalier Universitaire de Montpellier (eDOL, Montpellier, France). Adjudicated medical and pharmacy claims from the US (PharMetrics® Plus for Academics[28], US) and national health insurance claims from Estonia (CORIVA, Estonia). Finally, linked nationwide healthcare registries from Norway (NLHR@UiO[29], Norway) covering both primary and secondary care were included.

All contributing databases were previously mapped to the Observational Medical Outcomes Partnership (OMOP) Common Data Model, allowing for federated network analyses without sharing patient-level data. More detail on the participating databases can be found in Supplementary Tables 1 to 3.

Study population

All individuals registered in the participating databases for at least 365 days, and with a recorded COVID-19 diagnosis or positive SARS-CoV-2 test (polymerase chain reaction (PCR) or Nasopharyngeal Lateral Flow Assay rapid antigen test (TAR)) after 01/09/2020 comprised the source population. People with COVID-19 recorded before study start were excluded, as well as people with influenza in the 90 days before inclusion. Participants were excluded if they had a follow-up of < 120 days. Furthermore, we censored patients on the end of their observation period, death or end of COVID-19 testing in their country, if applicable.

Among those, we included people with any of the pre-defined 25 symptoms listed by the WHO clinical case definition for PCC[8], including: abdominal pain, allergy, altered smell and/or taste, anxiety, blurred vision, chest pain, cognitive dysfunction, cough, depression, dizziness, dyspnoea, fatigue or malaise, gastrointestinal issues (acid reflux, constipation, or diarrhoea), headache, intermittent fever, joint pain, memory issues, menstrual problems, muscles spasms or pain, neuralgia, pins and needles sensation, sleep disorder, tachycardia, post-exertional fatigue and tinnitus and hearing problems. Code lists and computable phenotypes were developed for each symptom and reviewed by 3 clinical experts from PCC clinics[30].

All individuals with at least one of those 25 symptoms recorded between 90 and 365 days after the date of COVID-19 diagnosis or SARS-CoV-2 positive test, and no record of that same symptom 180 days before COVID-19 were identified as patients with PCC and included for analysis.

Secondary analyses were conducted among people with multiple post-acute symptoms (≥ 2, and ≥ 3 symptoms) recorded following SARS-CoV-2 infection to ascertain a narrower definition of PCC.

In Supplementary Fig. 1 the attrition for all participating databases can be inspected.

Statistical Analyses

Descriptive analyses

Among all people with at least one post-acute COVID-19 symptom, key characteristics including age, sex and the number of different symptoms per person were summarised, and the frequency of each individual post-acute COVID-19

symptom calculated.

Clustering analyses

To retrieve clusters of associated symptoms, we performed LCA among all individuals with symptoms selected from the study databases (i.e. people with at least 1 post-acute COVID-19 symptom). LCA is a soft clustering algorithm, which calculates probabilities of belonging to different clusters and assigns patients to the one they are most likely to belong to, based on posterior probabilities[31,32].

We used LCA to derive clusters based on symptoms occurrence, considering sex and age as additional covariates. We created binary variables for each of the symptoms in question among all the patients belonging to the post-acute COVID-19 cohorts.

We ran the clustering algorithm with 100 random initialisations, initially setting the number of latent classes between 2 and 7. We selected the optimal number of classes based on Bayesian information criterion (i.e. BIC or CAIC), classification accuracy (i.e. mean posterior probability) and clinical relevance of the cluster characteristics, favouring solutions with fewer classes where possible[33,34]. We implemented the algorithm using the poLCA package[35] in R, v. 4.2.1.

We chose the LCA algorithm as it allows the testing of different hypotheses for the distribution of the data using goodness of fit tests, while accounting for covariates in the model[36,37].

Summary characteristics for clusters

Each cluster retrieved by LCA describes a specific subgroup of people with different symptoms. To better understand the baseline characteristics of those subgroups, we report demographics, vaccination status, selected co-morbidities, as well as healthcare utilisation (outpatient visits and hospitalisations) in the year before and the month after SARS-CoV-2 infection stratified by cluster.

Secondary analyses

We repeated cluster analyses among a subgroup of people with multiple co-existing symptoms (≥ 2 and ≥ 3 symptoms) for all databases that had sufficiently large sample sizes (i.e. ≥500 patients) in the final cohorts to be clustered.

In an additional sensitivity analysis, a record of “post-acute COVID-19” (OMOP Extension 705076) was included as an additional covariate to explore its correlation with individual symptoms.

We wrote the analytical code for this study in R version 4.2.3 and is available at [https://github.com/oxford-pharmacoepi/LongCovidStudyathon_W3]. All analyses were conducted separately for each database in a distributed manner and compiled by K.L. for this manuscript.

Declarations

Data availability

This study is based in part on data from the Clinical Practice Research Datalink (CPRD) obtained under license from the UK Medicines and Healthcare Products Regulatory Agency. We thank the patients who provided these data and the NHS who collected the data as part of their care and support.

Use of Clinical Practice Research Datalink (CPRD) data for this study was approved via the Research Data Governance (RDG) Process of the UK Medicines and Healthcare Products Regulatory Agency (protocol 23_002603).

Ethical approval for NHR@UiO in this study was obtained from The Regional Committee for Research Ethics (approval number 155294) and the Data Protection Officer at the University of Oslo (approval number 523275).

Ethical approval for CORIVA data was obtained from the Research Ethics Committee of the University of Tartu (No. 351/M-8).

Ethical epproval for HULAFE was obtained from the Comité de ética de la investigación con medicamentos (number 2023-232-1).

Ethical approval for IMASIS was obtained by the Parc de Salut Mar Research Ethics Committee CEIm-Parc de Salut Mar (number 2021/9975).

Ethical approval for IPCI was obtained by the Integrated Primary Care Information review board (registration number 9/2023).

For eDOL, no ethical approval was required according to French law for this study. All patients admitted to the hospital are provided with general information about the collection and secondary use of their data, and an opt-out option is offered.

PharMetrics® Plus for Academics needed no approval for use of pseudoanonymised secondary data.

Funding

This work has been funded by the European Health Data Evidence Network (EHDEN) through an Evidence Generation Fund Grant. EHDEN has received funding from the Innovative Medicines Initiative 2 Joint Undertaking under grant agreement No 806968. This Joint Undertaking receives support from the European Union's Horizon 2020 research and innovation programme and EFPIA.

The project was supported by the National Institute for Health and Care Research (NIHR) Oxford Biomedical Research Centre (BRC).

D.P.A. is funded through a NIHR Senior Research Fellowship (Grant number SRF-2018-11-ST2-004). The views expressed in this publication are those of the author(s) and not necessarily those of NIHR or the Department of Health and Social Care.

K.L. was supported by the Medical Research Council (grant number MR/W006731/1) and Bayer AG.

R.K. was supported by the European Regional Development Fund (RITA 1/02-120) and the Estonian Research Council grant (PRG1844).

L.P. was supported by a Sara Borrell fellowship awarded by the Spanish Institute of Health Carlos III (CD23/00223).

Author contributions

K.L, M.C, D.P.A, J.X and A.M.J. led the conceptualisation of the study.

K.K., D.P.A. and A.M.J. led the phenotyping of long COVID symptoms.

A.D. mapped and curated CPRD data.

K.L. and M.C. wrote the analytical code.

K.L., M.C., D.Ded., R.K., A.M., M.M., Z.C., D.Del., C.K., J.K., J.M., N.M., J.M.R., N.TH.T. conducted the statistical analyses on the respective databases.

D.Ded., T.D., R.L., G.M., A.A., J.T.A., T.B., E.B., S.K., K.K., C.L., M.A.M., A.N., H.ME.N., J.O.O., R.P., L.P.C., A.U., B.V., D.P.A., J.X., L.M. and A.M.J. clinically interpreted the results.

K.L., D.P.A., L.M. and A.M.J. wrote the first draft of the paper.

All authors read, contributed to, and approved the last version of the paper.

D.P.A and A.M.J. obtained the funding for this project.

Competing interests

D.P.A.'s department has received grant/s from Amgen, Chiesi–Taylor, Lilly, Janssen, Novartis, and UCB Biopharma. His research group has received consultancy fees from Astra Zeneca and UCB Biopharma. Amgen, Astellas, Janssen, Synapse Management Partners and UCB Biopharma have funded or supported training programmes organised by DPA's department. R.P. reports serving on advisory boards for Gilead Sciences, Inc, Pfizer, Inc, Roche Therapeutics, MSD, GSK, ViiV Healthcare, Eli Lilly and Company, PharmaMar, and Atea Pharmaceuticals, Inc; and receiving research grants paid to his institution from MSD, ViiV Healthcare, Gilead Sciences, and PharmaMar. L.M. reports receiving grants from Grifols; receiving honoraria as a speaker from AstraZeneca, Gilead Sciences, GSK, and Pfizer; and participation in advisory boards for Gilead Sciences and Merck. M.M. works for a research group that in the past 3 years received unconditional research grants from Chiesi, UCB, Amgen, Johnson & Johnson, Innovative Medicines Initiative and the European Medicines Agency. D.Ded., Z.C. and J.O. are employees of the Medicines and Healthcare Products Regulatory Agency, which provides the CPRD research service. K.K. is a consortial author in the US National Institutes of Health National COVID Cohort Collaborative (funding expired in 2022 with no renewal or active impact on any current work).

References

1. Lopez-Leon S, Wegman-Ostrosky T, Perelman C, et al. More Than 50 Long-Term Effects of COVID-19: A Systematic Review and Meta-Analysis. *Res Sq* 2021 doi: 10.21203/rs.3.rs-266574/v1 [published Online First: 2021/03/11]
2. Nalbandian A, Sehgal K, Gupta A, et al. Post-acute COVID-19 syndrome. *Nat Med* 2021 doi: 10.1038/s41591-021-01283-z [published Online First: 2021/03/24]
3. Greenhalgh T, Knight M, A'Court C, et al. Management of post-acute covid-19 in primary care. *BMJ* 2020;370:m3026. doi: 10.1136/bmj.m3026 [published Online First: 2020/08/14]
4. Whitaker M, Elliott J, Chadeau-Hyam M, et al. Persistent COVID-19 symptoms in a community study of 606,434 people in England. *Nat Commun* 2022;13(1):1957. doi: 10.1038/s41467-022-29521-z [published Online First: 2022/04/14]
5. Subramanian A, Nirantharakumar K, Hughes S, et al. Symptoms and risk factors for long COVID in non-hospitalized adults. *Nat Med* 2022;28(8):1706-14. doi: 10.1038/s41591-022-01909-w [published Online First: 2022/07/26]
6. Bowe B, Xie Y, Al-Aly Z. Postacute sequelae of COVID-19 at 2 years. *Nat Med* 2023 doi: 10.1038/s41591-023-02521-2 [published Online First: 2023/08/22]

7. The Lancet, Long COVID: 3 years in, The Lancet, Volume 401, Issue 10379, 2023, Page 795, ISSN 0140-6736, https://doi.org/10.1016/S0140-6736(23)00493-2.
8. University CaJH. COVID-19 Dashboard https://coronavirus.jhu.edu/map.html2023 [March 2023:[
9. WHO. A clinical case definition of post COVID-19 condition by a Delphi consensus 2021 [Available from: https://www.who.int/publications/i/item/WHO-2019-nCoV-Post_COVID-19_condition-Clinical_case_definition-2021.1 accessed 6 October 2021.
10. Gottlieb M, Spatz ES, Yu H, et al. Long COVID Clinical Phenotypes up to 6 Months After Infection Identified by Latent Class Analysis of Self-Reported Symptoms. *Open Forum Infect Dis* 2023;10(7):ofad277. doi: 10.1093/ofid/ofad277 [published Online First: 2023/07/10]
11. Tsuchida T, Yoshimura N, Ishizuka K, et al. Five cluster classifications of long COVID and their background factors: A cross-sectional study in Japan. *Clin Exp Med* 2023;23(7):3663-70. doi: 10.1007/s10238-023-01057-6 [published Online First: 2023/04/08]
12. Kenny G, McCann K, O'Brien C, et al. Identification of Distinct Long COVID Clinical Phenotypes Through Cluster Analysis of Self-Reported Symptoms. *Open Forum Infect Dis* 2022;9(4):ofac060. doi: 10.1093/ofid/ofac060 [published Online First: 2022/03/11]
13. Fischer A, Badier N, Zhang L, et al. Long COVID Classification: Findings from a Clustering Analysis in the Predi-COVID Cohort Study. *Int J Environ Res Public Health* 2022;19(23) doi: 10.3390/ijerph192316018 [published Online First: 2022/12/12]
14. Català M, Mercadé-Besora N, Kolde R, et al. The effectiveness of COVID-19 vaccines to prevent long COVID symptoms: staggered cohort study of data from the UK, Spain, and Estonia. Lancet Respir Med. 2024 Jan 11:S2213-2600(23)00414-9. doi: 10.1016/S2213-2600(23)00414-9.
15. Global Burden of Disease Long CC, Wulf Hanson S, Abbafati C, et al. Estimated Global Proportions of Individuals With Persistent Fatigue, Cognitive, and Respiratory Symptom Clusters Following Symptomatic COVID-19 in 2020 and 2021. *JAMA* 2022;328(16):1604-15. doi: 10.1001/jama.2022.18931 [published Online First: 2022/10/11]
16. Fernandez-de-Las-Penas C, Martin-Guerrero JD, Florencio LL, et al. Clustering analysis reveals different profiles associating long-term post-COVID symptoms, COVID-19 symptoms at hospital admission and previous medical co-morbidities in previously hospitalized COVID-19 survivors. *Infection* 2023;51(1):61-69. doi: 10.1007/s15010-022-01822-x [published Online First: 2022/04/23]
17. Reese JT, Blau H, Casiraghi E, et al. Generalisable long COVID subtypes: findings from the NIH N3C and RECOVER programmes. *EBioMedicine* 2023;87:104413. doi: 10.1016/j.ebiom.2022.104413 [published Online First: 2022/12/24]
18. Zhang H, Zang C, Xu Z, et al. Data-driven identification of post-acute SARS-CoV-2 infection subphenotypes. *Nat Med* 2023;29(1):226-35. doi: 10.1038/s41591-022-02116-3 [published Online First: 2022/12/02]
19. Mateu, Lourdes et al., Determinants of the onset and prognosis of the post-COVID-19 condition: a 2-year prospective observational cohort study, The Lancet Regional Health – Europe, Volume 33, 100724
20. Thaweethai T, Jolley SE, Karlson EW, et al. Development of a Definition of Postacute Sequelae of SARS-CoV-2 Infection. *JAMA.* 2023;329(22):1934–1946. doi:10.1001/jama.2023.8823
21. Ryan, P et al. The Book of OHDSI: Chapter 4 - The Common Data Model [Available from: https://ohdsi.github.io/TheBookOfOhdsi/CommonDataModel.html.
22. Burn E, Fernández-Bertolín S, Voss EA, Blacketer C, Aragón M, Recalde M, et al. Establishing and characterising large COVID-19 cohorts after mapping the Information System for Research in Primary Care in Catalonia to the OMOP Common Data Model. medRxiv. 2021.

23. Gandjour A. Long COVID: Costs for the German economy and health care and pension system. BMC Health Serv Res. 2023;23(1):641. Published 2023 Jun 14. doi:10.1186/s12913-023-09601-6
24. Vlug A, van der Lei J, Mosseveld B, et al. Postmarketing surveillance based on electronic patient records: the IPCI project. *Methods of information in medicine* 1999;38:339-44.
25. Herrett E, Gallagher AM, Bhaskaran K, et al. Data Resource Profile: Clinical Practice Research Datalink (CPRD). *Int J Epidemiol* 2015;44(3):827-36. doi: 10.1093/ije/dyv098 [published Online First: 2015/06/08]
26. Wolf A, Dedman D, Campbell J, et al. Data resource profile: Clinical Practice Research Datalink (CPRD) Aurum. *Int J Epidemiol* 2019;48(6):1740-40g. doi: 10.1093/ije/dyz034 [published Online First: 2019/03/13]
27. JJ Sancho IP, D Domènech, M Martín-Baranera, J Palau, F Sanz. IMASIS. A Multicenter Hospital Information System - Experience in Barcelona. *Stud Health Technol Inform* 1998;56:35 - 42. doi: 10.3233/978-1-60750-900-4-35
28. IQVIA. IQVIA PharMetrics Plus for Academics Enhanced with Mortality Data https://www.iqvia.com/locations/united-states/library/fact-sheets/iqvia-pharmetrics-plus-for-academics-enhanced-with-mortality-data2023 [
29. Bakken IJ, Ariansen AMS, Knudsen GP, et al. The Norwegian Patient Registry and the Norwegian Registry for Primary Health Care: Research potential of two nationwide health-care registries. *Scand J Public Health* 2020;48(1):49-55. doi: 10.1177/1403494819859737 [published Online First: 20190709]
30. Kostka K, Roel E, Trinh N, et al. The burden of long COVID: a multinational cohort analysis of Spanish and UK data including SARS-CoV-2 infections, reinfections, and matched contemporaneous test negative controls. *Research Square* 2023 doi: 10.21203/rs.3.rs-2642600/v1
31. Lanza ST, Rhoades BL. Latent class analysis: an alternative perspective on subgroup analysis in prevention and treatment. *Prev Sci* 2013;14(2):157-68. doi: 10.1007/s11121-011-0201-1
32. Naldi L, Cazzaniga S. Research Techniques Made Simple: Latent Class Analysis. *J Invest Dermatol* 2020;140(9):1676-80 e1. doi: 10.1016/j.jid.2020.05.079
33. Bozdogan H. Model selection and Akaike's information criterion (AIC): The general theory and its analytical extensions. *Psychometrika* 1987;52(3):345-70. doi: https://doi.org/10.1007/BF02294361
34. Weller BE, Bowen NK, Faubert SJ. Latent Class Analysis: A Guide to Best Practice. *Journal of Black Psychology* 2020;46(4):287-311. doi: 10.1177/0095798420930932
35. Drew A. Linzer JBL. poLCA: An R Package for Polytomous Variable Latent Class Analysis. *Journal of Statistical Software* 2011;42(10):1-29. doi: 10.18637/jss.v042.i10
36. Feuillet F, Bellanger L, Hardouin JB, et al. On Comparison of Clustering Methods for Pharmacoepidemiological Data. *J Biopharm Stat* 2015;25(4):843-56. doi: 10.1080/10543406.2014.920855
37. Goodman LA. Latent Class Analysis: The Empirical Study of Latent Types, Latent Variables, and Latent Structures. Applied Latent Class Analysis2002:3-55.

Figures

Figure 1

Post-acute COVID-19 symptoms and OMOP Extensionpost-COVID condition recorded after COVID-19 infection per database. The concept “OMOP Extensionpost-COVID-condition diagnosis" refers to the OMOP Extension code 705076. CPRD stands for Clinical Practice Research Datalink, IPCI for Integrated Primary Care Information, HULAFE for The Health Research Institute Hospital La Fe, IMASIS for Parc de Salut Mar Barcelona Information System and NLHR@UiO stands for Norwegian Linked Health Registries at University of Oslo.

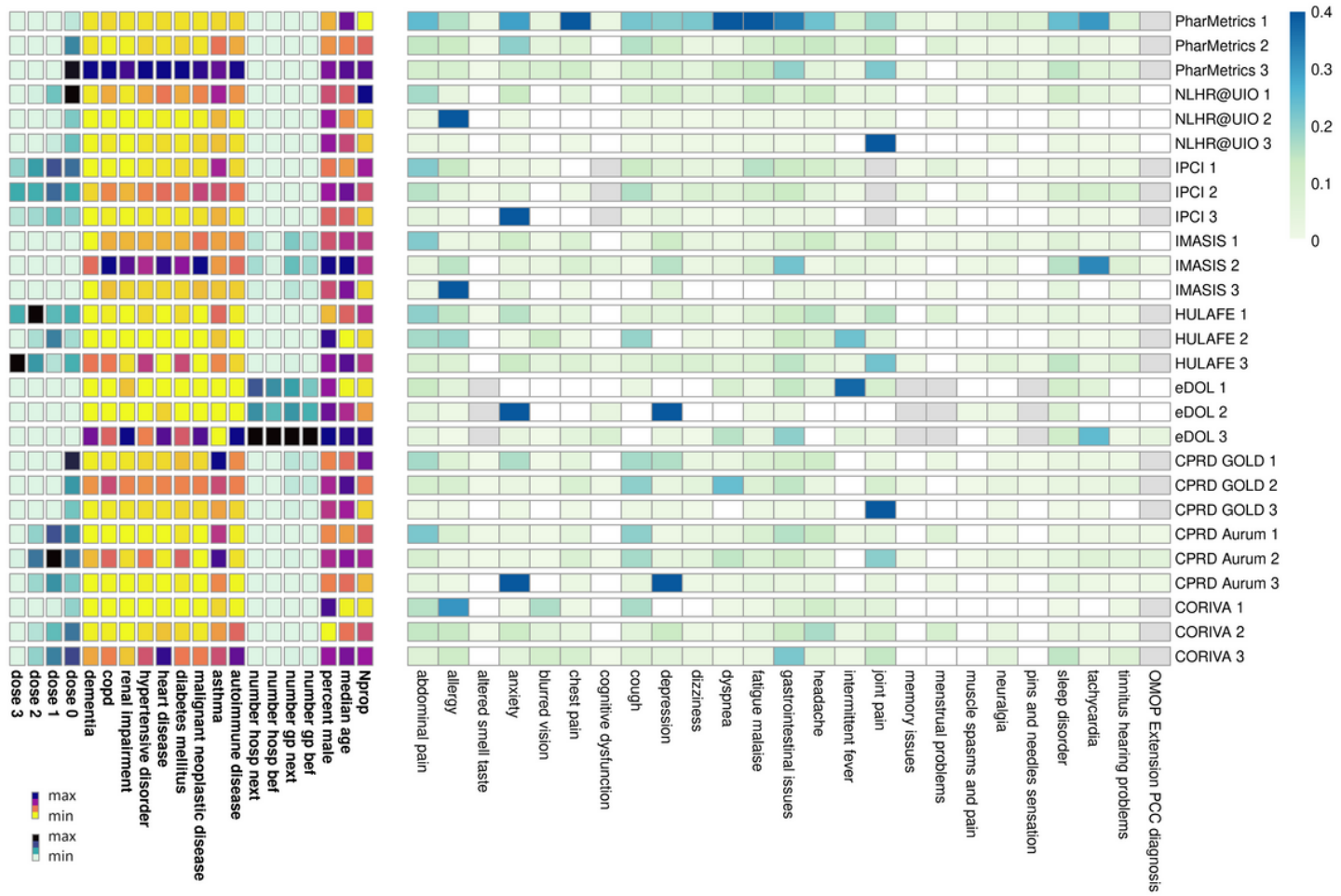

Figure 2

Proportion of symptoms per cluster (right) and their characteristics (left) for the contributing databases, for the cohorts of people with one or more symptoms post infection.

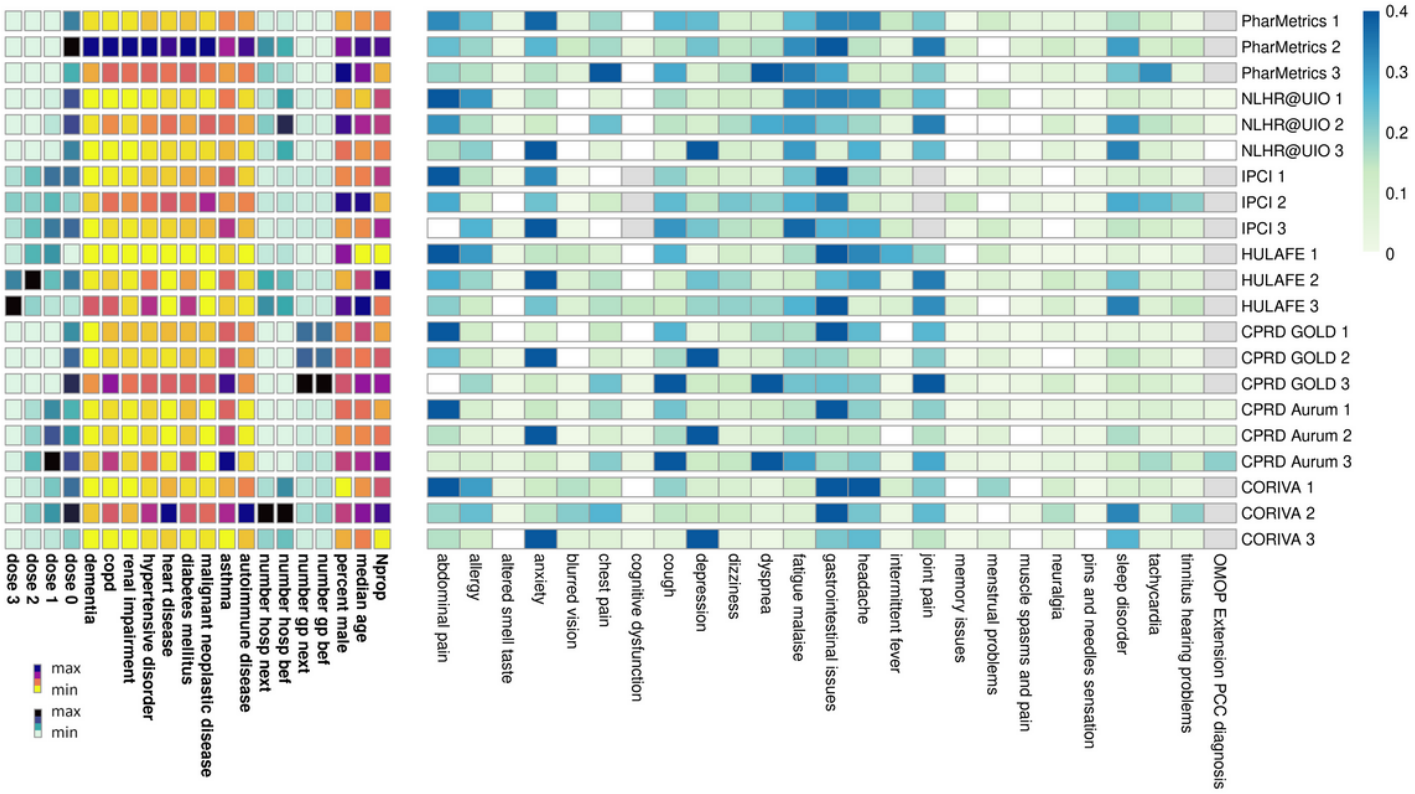

Figure 3

Proportion of symptoms per cluster (right) and their characteristics (left) for the contributing databases, for the cohorts of people with three or more symptoms post infection.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SupplementaryPatternpostacuteCOVID19conditions1.docx